



COMP9444: Neural Networks and Deep Learning

Week 1b. Neuroanatomy

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What is a Neural Network?

- massively parallel distributed processor made up of simple processing units
- knowledge acquired from environment through a learning process
- knowledge stored in the form of synaptic weights

Why Neural Networks?

- biologically inspired
- good learning properties
- continuous, nonlinear
- well adapted to certain tasks
- fault tolerant
- graceful degradation

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Sub-Symbolic Processing



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Dot Illusion



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Theories about Intelligence

- 380BC Plato (Rationalism - innateness)
- 330BC Aristotle (Empiricism - experience)
- 1641 Descartes (mind-body Dualism)
- 1781 Kant (Critique of Pure Reason)
- 1899 Sigmund Freud (Psychology)
- 1953 B.F. Skinner (Behaviourism)

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Artificial Intelligence Origins

- 1642 Blaise Pascal (mechanical adding machine)
- 1694 Gottfried Leibniz (mechanical calculator)
- 1769 Wolfgang von Kempelen (Mechanical Turk)
- 1837 Charles Babbage & Ada Lovelace (Difference Engine)
- 1848 George Boole (the Calculus of Logic)
- 1879 Gottlob Frege (Predicate Logic)
- 1950 Turing Test
- 1956 Dartmouth conference

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Neural Network Origins

- 1943 McCulloch & Pitts (neuron models)
- 1948 Norbert Wiener (Cybernetics)
- 1948 Alan Turing (B-Type Networks)
- 1955 Oliver Selfridge (Pattern Recognition)
- 1962 Hubel and Wiesel (visual cortex)
- 1962 Frank Rosenblatt (Perceptron)

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Serial Symbolic AI

- 1956 Newell & Simon (Logic Theorist)
- 1959 John McCarthy (Lisp)
- 1959 Arthur Samuel (Checkers)
- 1965 Joseph Weizenbaum (ELIZA)
- 1967 Edward Feigenbaum (Dendral)

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Neural Network “Dark Ages”

- 1969 Minsky & Papert published “Perceptrons”, emphasizing the limitations of neural models, and lobbied agencies to cease funding neural network research.
- from 1969 to 1985, very little work in neural networks or machine learning.
- a few exceptions, e.g. Stephen Grossberg, Teuvo Kohonen, Paul Werbos.

Knowledge-Based Systems

- 1970's and early 1980's, AI research focused on symbolic processing, Expert Systems
- Some commercial success, but ran into difficulties:
 - combinatorial explosion in search spaces
 - difficulty of formalising everyday knowledge as well as expert knowledge

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Neural Network Renaissance

- 1986 Rumelhart, Hinton & Williams (multi-layer, backprop)
- 1989 Dean Pomerleau (ALVINN)
- late 1980's renewed enthusiasm, hype
- 1990's more principled approaches
- 2000's SVM, Bayesian models became more popular
- 2010's deep learning networks, GPU's
- 2020's transformers, stable diffusion

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Applications of Deep Learning

- | | |
|---|---|
| → Image Processing <ul style="list-style-type: none">→ Classification→ Segmentation→ Image Generation | → Game Playing <ul style="list-style-type: none">→ Deep Q-Learning→ AlphaGo→ AlphaStar |
| → Language Processing <ul style="list-style-type: none">→ Translation→ Sentiment Analysis→ Text Generation→ Chatbots | → Combining Images and Language <ul style="list-style-type: none">→ Automatic Captioning→ Generating Images from Text→ Video Generation→ Multimodal AI |

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History of Deep Learning

Two perspectives on the history of Deep Learning

Viewpoint 1: Focusing on recent work (after 2012)

<https://www.cs.toronto.edu/~hinton/absps/NatureDeepReview.pdf>

Viewpoint 2: Focusing on earlier work (before 2012)

<http://people.idsia.ch/~juergen/deep-learning-overview.html>

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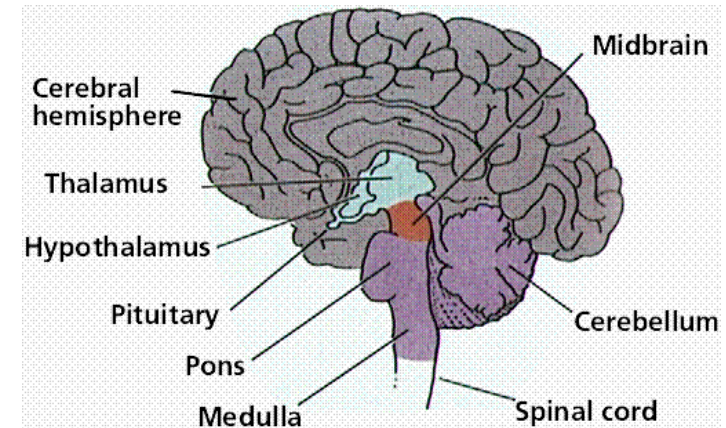
Neuroanatomy

- Central Nervous System
 - Brain
 - Spinal cord
- Peripheral Nervous System
 - Somatic nervous system
 - Autonomic nervous system
 - Enteric nervous system

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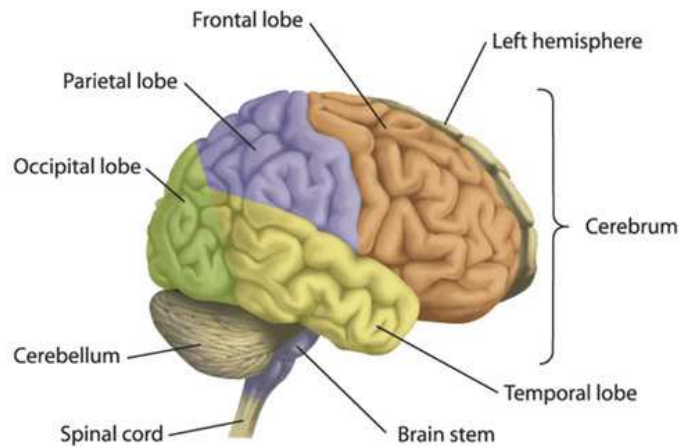
Brain Regions



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Cerebral Cortex



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Cerebral Cortex

- “cortex” from Latin word for “bark” (of tree)
- cortex is a sheet of tissue making up outer layers of brain, 2-6cm thick
- right and left sides connected by corpus callosum
- functions: thought, voluntary movement, language, reasoning, perception

Brain Stem

- general term for area of brain between the thalamus and spinal cord
- includes medulla, pons, tectum, reticular formation and tegmentum
- functions: breathing, heart rate, blood pressure, and others

Cerebellum

- from Latin word for “little brain”
- functions: movement, balance, posture

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Midbrain

- functions: vision, audition, eye movement, body movement

Thalamus

- receives sensory information and relays it to the cerebral cortex
- also relays information from the cerebral cortex to other areas of the brain, and the spinal cord
- functions: sensory integration, motor integration

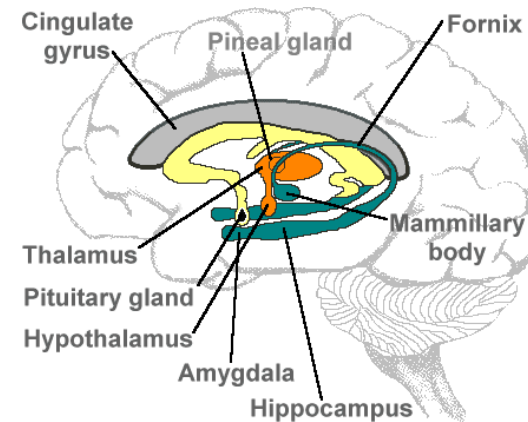
Hypothalamus

- composed of several different areas at the base of the brain
- the size of a pea (about 1/300 of the total brain weight)
- functions: body temperature, emotions, hunger, thirst, circadian rhythms

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Limbic System



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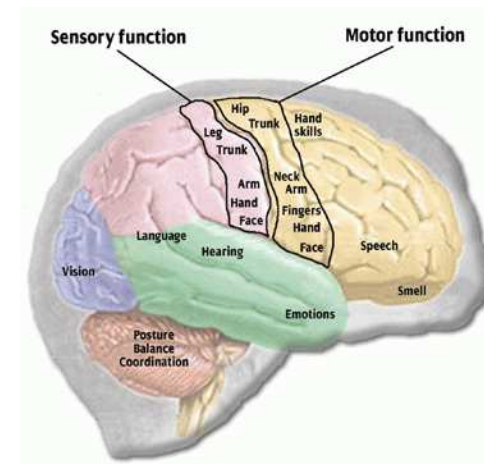
Limbic System

- group of structures including amygdala, hippocampus, mammillary bodies and cingulate gyrus
- important for controlling the emotional response to a given situation
- hippocampus also important for memory
- functions: emotional behaviour

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Brain Functions



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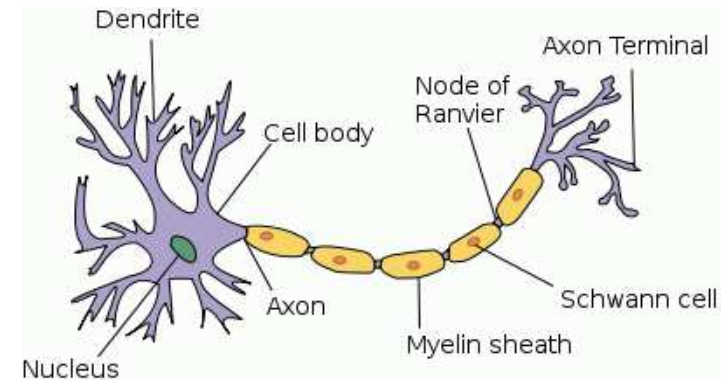
Neurons as Body Cells

- The body is made up of billions of cells. Cells of the nervous system, called *neurons*, are specialized to carry “messages” through an electrochemical process.
- The human brain has about 100 billion neurons, and a similar number of support cells called “glia”.
- Neurons are similar to other cells in the body in some ways, such as:
 - neurons are surrounded by a cell membrane
 - neurons have a nucleus that contains genes (DNA)
 - neurons carry out basic cellular processes like protein synthesis and energy production

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Structure of a Typical Neuron



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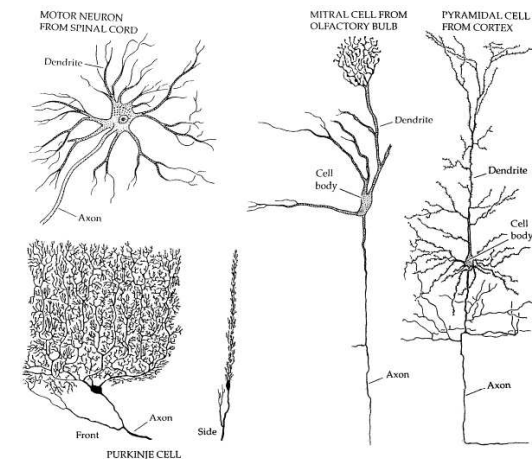
Neurons versus Body Cells

- Neurons have specialized extensions called *dendrites* and *axons*. Dendrites bring information to the cell body, while axons take information away from the cell body.
- The axon of one neuron can connect to the dendrite of another neuron through an electrochemical junction called a *synapse*.
- Most neurons have only one axon, but the number of dendrites can vary widely:
 - Unipolar and Bipolar neurons have only one dendrite
 - Purkinje neurons can have up to 100,000 dendrites

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Variety of Neuron Types



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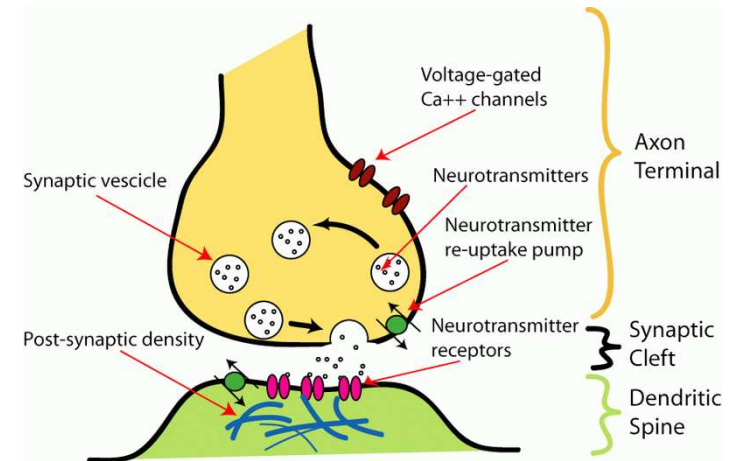
Axons and Dendrites

- Dendrites are typically less than a millimetre in length
- Axons can vary in length from less than a millimetre to more than a metre (motor neurons)
- Long axons are sometimes surrounded by a myelinated sheath, which prevents the electrical signal from dispersing, and allows it to travel faster (up to 100 m/s).

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Synapse



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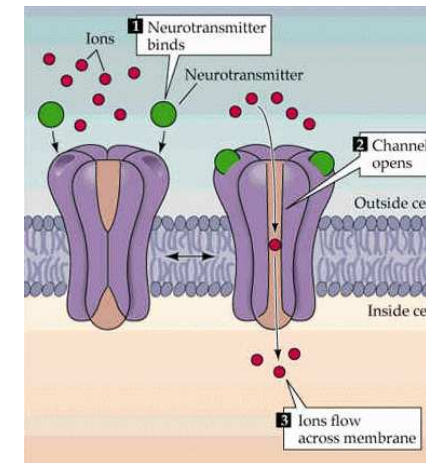
Synapses and Ion Channels

- electrical pulse reaches the endbulb and causes the release of neurotransmitter molecules from little packets (vesicles) through the synaptic membrane
- transmitter then diffuses through the synaptic cleft to the other side
- when the neurotransmitter reaches the post-synaptic membrane, it causes a change in polarisation of the membrane
- the change in potential can be *excitatory* (moving the potential towards the threshold) or *inhibitory* (moving it away from the threshold)

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Ion Channel



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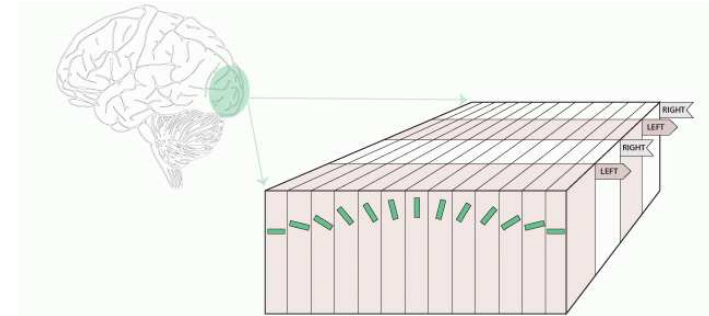
The Big Picture

- human brain has 100 billion neurons with an average of 10,000 synapses each
- latency is about 3-6 milliseconds
- therefore, at most a few hundred “steps” in any mental computation, but massively parallel

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Hubel and Weisel – Visual Cortex

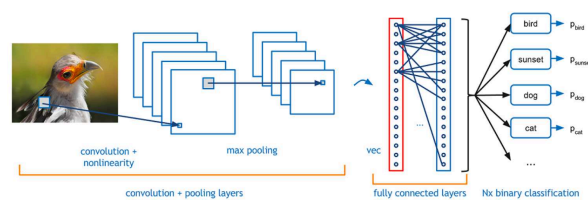


- cells in the visual cortex respond to lines at different angles
- cells in V2 respond to more sophisticated visual features
- Convolutional Neural Networks are inspired by this neuroanatomy
- CNN's can now be simulated with massive parallelism, using GPU's

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Convolutional Networks

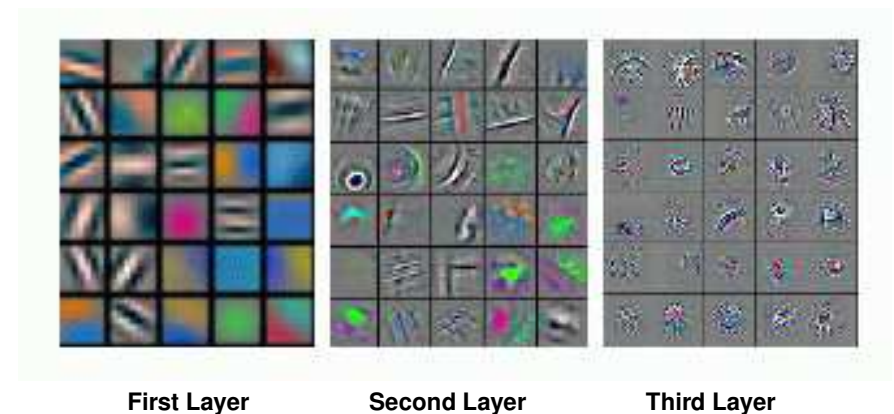


Suppose we want to classify an image as a bird, sunset, dog, cat, etc.
 If we can identify features such as feather, eye, or beak which provide useful information in one part of the image, then those features are likely to also be relevant in another part of the image.
 We can exploit this regularity by using a convolution layer which applies the same weights to different parts of the image.

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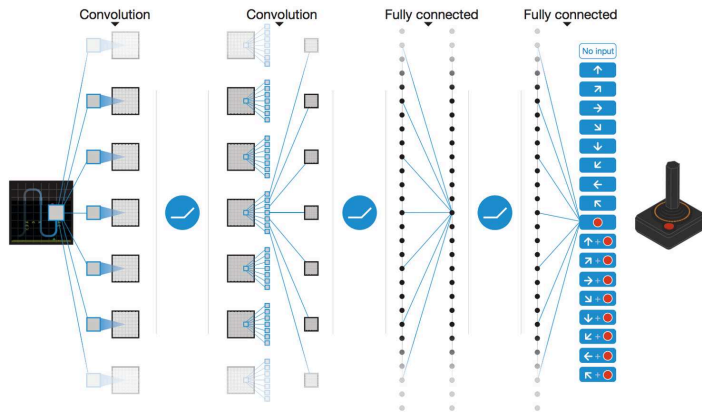
Convolutional Filters



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Deep Q-Learning



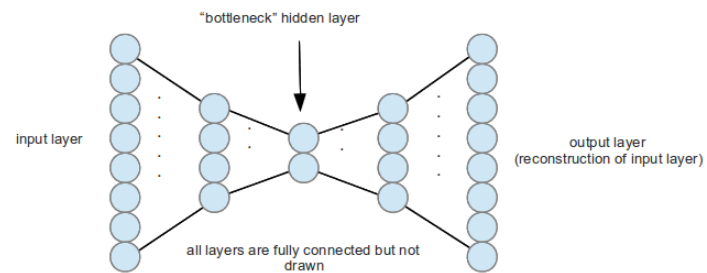
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Starcraft



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Autoencoder Networks



- output is trained to reproduce the input as closely as possible
- activations normally pass through a bottleneck, so the network is forced to compress the data in some way

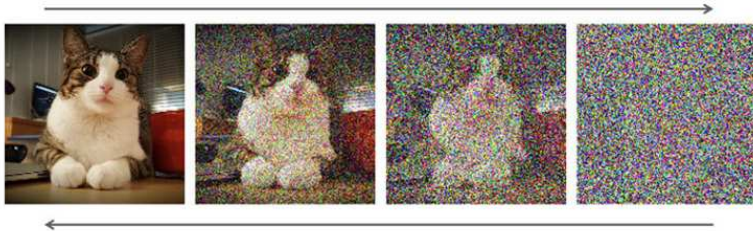
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Generative Adversarial Networks

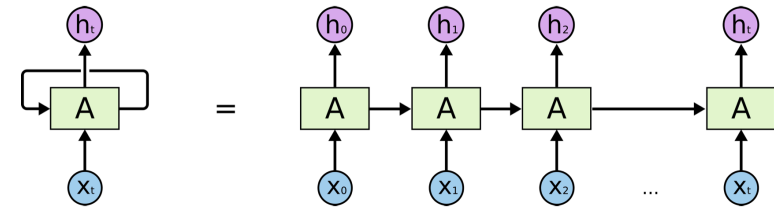


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Stable Diffusion



Recurrent Neural Networks

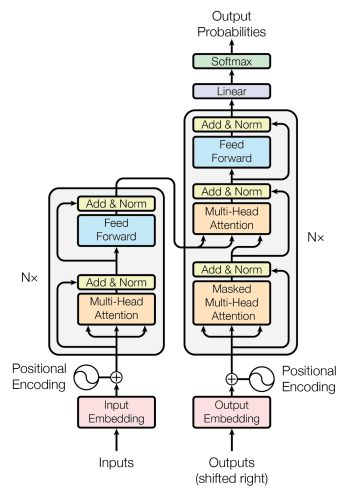


- can “unroll” a recurrent architecture into an equivalent feedforward architecture, with shared weights
- useful for processing language or other temporal sequences

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Transformer



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